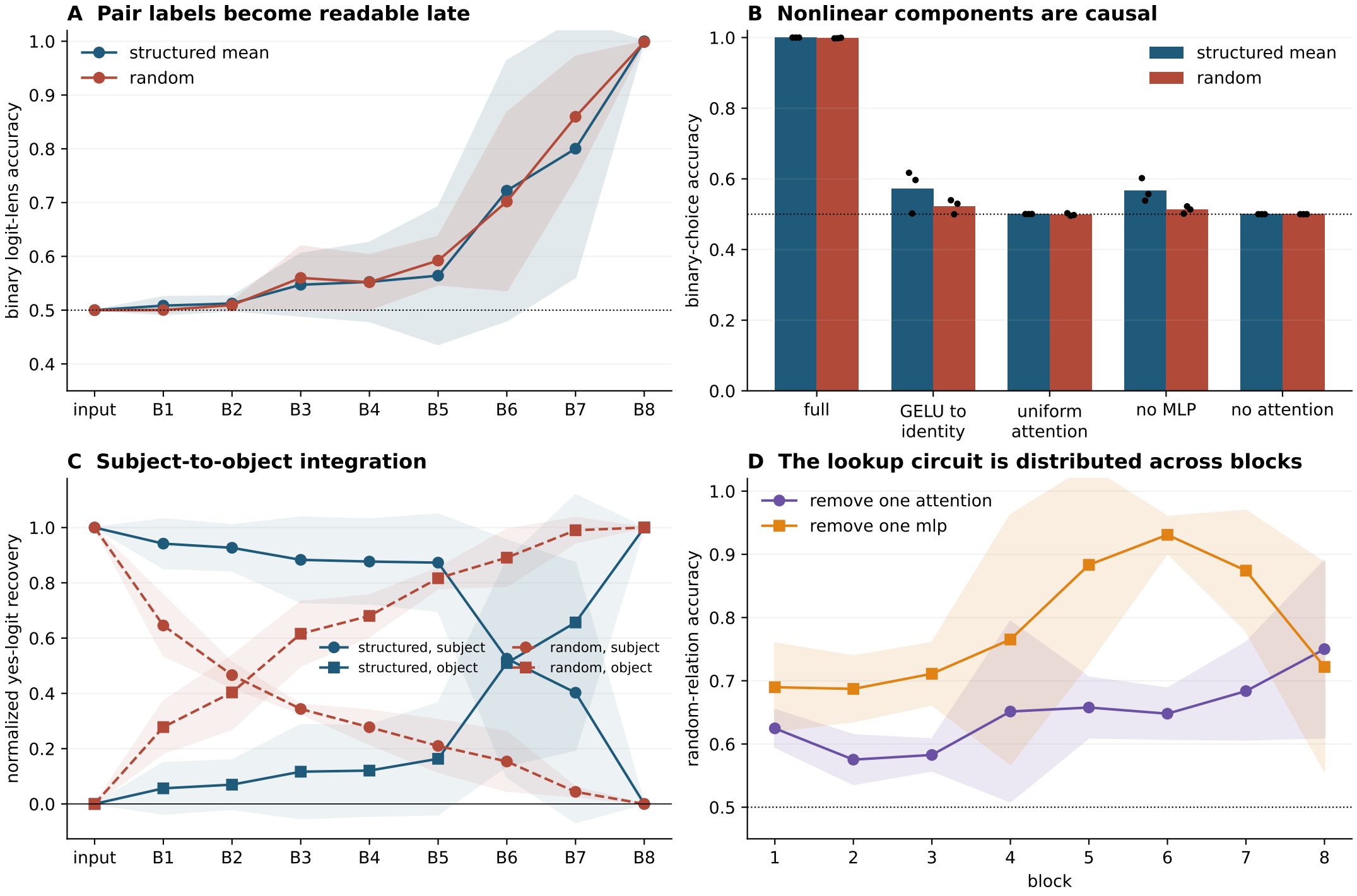
A Pair labels become readable late
structured mean
random
binary logit-lens accuracy
1.0
0.9
0.8
0.7
0.6
0.5
0.4
input
B1
B2
B3
B4
B5
B6
B7
B8
B Nonlinear components are causal
structured mean
random
binary-choice accuracy
1.0
0.8
0.6
0.4
0.2
0.0
full
GELU to identity
uniform attention
no MLP
no attention
C Subject-to-object integration
structured, subject
structured, object
random, subject
random, object
normalized yes-logit recovery
1.0
0.8
0.6
0.4
0.2
0.0
input
B1
B2
B3
B4
B5
B6
B7
B8
D The lookup circuit is distributed across blocks
remove one attention
remove one mlp
random-relation accuracy
1.0
0.9
0.8
0.7
0.6
0.5
1
2
3
4
5
6
7
8
block